Agent-Based Simulation of Organisational Misconduct

Reporting, Enforcement, and Hidden Misconduct

Presentation of Group 5

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1. Recap: Research Question and Output

Research Question

Under what conditions does stricter enforcement reduce true misconduct

We are interested in the gap between:

Visible / Punished misconduct and total misconduct in the organisation.

Output	Meaning
True misconduct	Total amount of misconduct actually occurring in the organization
Hidden misconduct	Misconduct that occurs but is not sanctioned or detected
Hidden Misconduct Rate	= (True Misconduct - Sanctioned Misconduct) / True Misconduct. **Captures the key question**: is the organization reducing misconduct itself, or only reducing what becomes visible?

3. Agents and Environment

Agents

One Agent Type:
Employees

They can both

- Commit misconduct
- Report misconduct

Environment

One room with a fixed population of employees

- Agents move randomly and interact locally through the observation of misconduct.
- Misconduct can only be reported and sanctioned if it is observed by nearby agents.
- Reporter protection determines Retaliation chance.
- Punishment value determines Retaliation strength.
- Misconduct propensity and fear evolve over time through learning and feedback mechanisms.

(Reductionistic)
Deliberately kept only the essential structure needed to model misconduct, reporting, and feedback

4. Agent Attributes

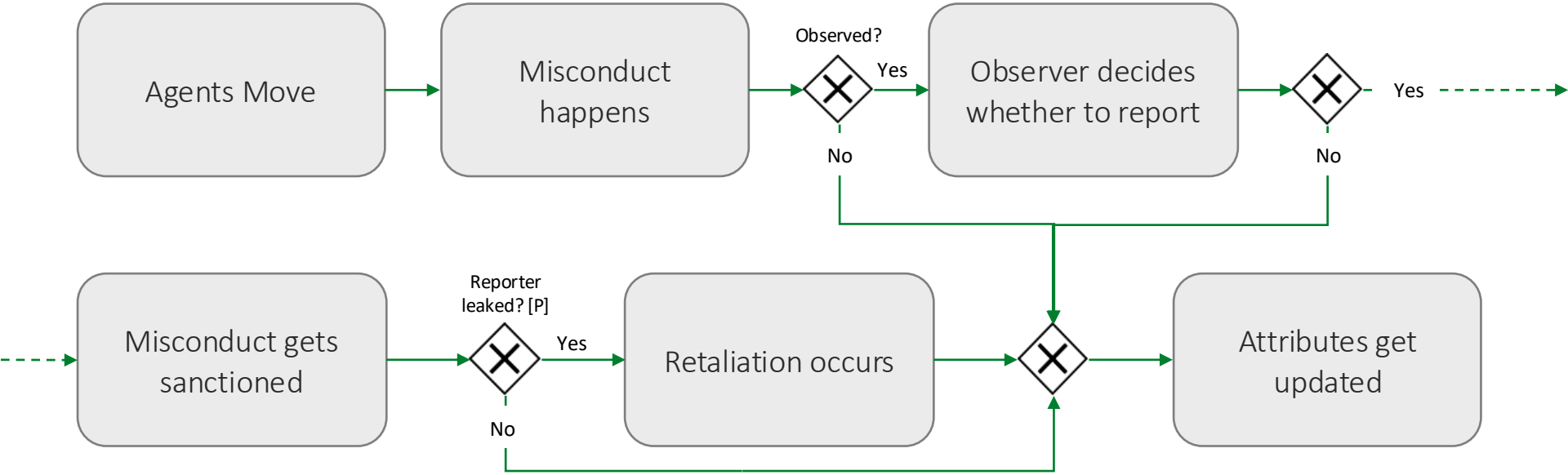
Attribute		Meaning
Employee Utility	U	Personal Utility attributed to each Employee. Increases by successful misconduct. Decreases by Punishment & Retaliation
Misconduct propensity	M	Likelihood that an employee engages in misconduct for personal benefit at any point in time. Decreases after punishment and adapts over time.
Fear of retaliation	F	Determines willingness to report misconduct. Increases after retaliation and decays gradually.
Location	L	Current location of the Agent. Determines whether misconduct can be observed by nearby agents.

5. Environmental Parameters (independent variables)

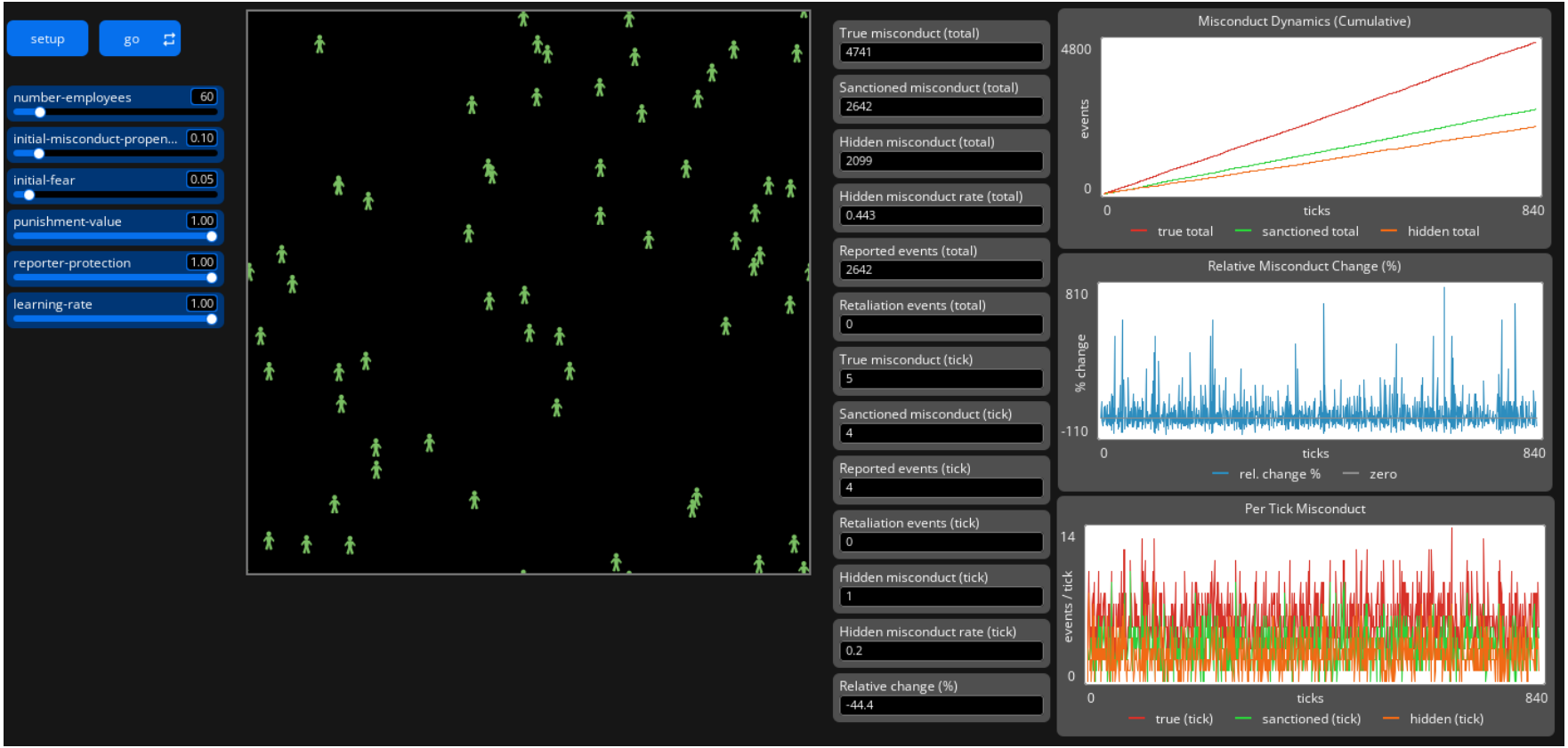
Parameter		Meaning
Number of employees	N	Determines interaction density and probability of observation
Initial misconduct propensity	M_0	Baseline tendency of agents to engage in misconduct
Initial fear	F_0	Initial reluctance to report misconduct
Punishment value	E	Strength of sanctioning effect (enforcement intensity)
Reporter protection	P	Reduces retaliation risk and increases reporting probability
Learning rate	λ	Speed at which agents adapt behaviour over time

6. What happens per Tick

Each simulation tick updates misconduct, reporting, investigation, and feedback.



7. Netlogo


setup
go
number-employees 60
initial-misconduct-propen... 0.10
initial-fear 0.05
punishment-value 1.00
reporter-protection 1.00
learning-rate 1.00
True misconduct (total) 4741
Sanctioned misconduct (total) 2642
Hidden misconduct (total) 2099
Hidden misconduct rate (total) 0.443
Reported events (total) 2642
Retaliation events (total) 0
True misconduct (tick) 5
Sanctioned misconduct (tick) 4
Reported events (tick) 4
Retaliation events (tick) 0
Hidden misconduct (tick) 1
Hidden misconduct rate (tick) 0.2
Relative change (%) -44.4
Misconduct Dynamics (Cumulative)
4800
events
0
0
ticks
840
true total
sanctioned total
hidden total
Relative Misconduct Change (%)
810
% change
-110
0
ticks
840
rel. change %
zero
Per Tick Misconduct
14
events / tick
0
0
ticks
840
true (tick)
sanctioned (tick)
hidden (tick)

8. Current Model State

Working

- ✓ Agent movement and local observation
- ✓ Probabilistic misconduct
- ✓ Logistic reporting function
- ✓ Auto-sanctioning, punishment-value
- ✓ Retaliation → fear spike (learning-rate coupled)
- ✓ Fear decay between retaliation events
- ✓ All 3 plots (cumulative, per-tick, relative %)

Reasonable, not final

- ~ Misconduct propensity values look plausible
- ~ Hidden misconduct rate responds to policy changes in expected direction
- ~ Fear decay works but decay speed may need calibration
- ~ Reporting formula constants (0.8, 0.2) not yet empirically grounded

Not yet implemented

- ✗ Expected utility function for reporting
- ✗ Detectability window (misconduct visible for N ticks)
- ✗ Reporter utility gain from reporting
- ✗ Compliance unit / manager agents

9. Next Steps

1 — Implement expected utility

Replace simple propensity with an expected-value function per agent. This allows heterogeneous histories to drive different behaviours.

2 — Detectability window

Misconduct stays detectable for N ticks after the event. Observers can still report it in subsequent ticks, making the timing of observation and reporting more realistic.

3 — Reporter utility gain

Reporters gain utility from reporting (moral benefit, financial incentive). Introduces a positive force countering fear — allowing the model to explore when incentive design matters more than protection.

4 — Social learning after reporting

An agent's own misconduct propensity adapts after reporting someone. Reporting may signal personal integrity (propensity drops) or trigger fear of retaliation (propensity rises as a protective response).

Thank you

Questions and feedback